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Cancer Surgery

Surgery has been used to treat cancer for many, many years. Surgery also plays a key role in diagnosing cancer and finding out how far it may have spread.

Cancer surgery is done by specialists called **surgical oncologists**. Cancer surgery is also done by other types of doctors who specialize in cancer or other types of surgery.

[How surgery is used for cancer](#)

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How surgery is used for cancer

Cancer surgery is done to remove part or all of a tumor (growth) from the body. Surgery was the first kind of cancer treatment developed and is used to treat many types of cancer.

Surgery is used for cancer in many ways.

To diagnose cancer

Figuring out if you have cancer can mean that some or all of a tumor will have to be removed from your body. This procedure is called a biopsy.

There are different kinds of biopsies. During a surgical biopsy, your doctor will make a small cut to remove a tissue sample. They may take a small part or all of a tumor. Other types of biopsies use needles instead of surgery, including image-guided fine needle aspiration and core biopsies.

The type of biopsy done depends on where the tumor is and what type of cancer it is thought to be. For example, the method used for prostate biopsies is different from those used for lung biopsies.

After a biopsy, a pathologist looks at the tissue under a microscope. A pathologist is a doctor who looks at cancer cells, tissues, and organs to find disease. The pathologist sends a report to your primary or oncology care team to let them know whether the tumor is cancer and, if so, what type.

Learn more about different types of biopsies in [Testing Biopsy and Cytology Specimens for Cancer](#)¹.

To stage cancer

Staging surgery is done to find out how large your tumor is, if it has spread, and if so, where.

Your doctor may remove the entire tumor or take just part of it. They may also take some lymph nodes near the tumor to find out if your cancer has spread there. Lymph nodes are usually small, bean-shaped organs that help fight infection. They are often the first place a cancer spreads.

A pathologist will look at the removed tissue and let your oncologist know what they found. The results of staging surgery and other tests help you and your doctor decide on the best treatment.

[Imaging](#)² tests may also be done to help stage your cancer. These might include tests such as ultrasounds, CT scans, MRIs, and X-rays.

To learn more about this, see [Cancer Staging](#)³.

To treat cancer

Many people with cancer will have surgery as part of their cancer treatment. Sometimes, surgery is the only cancer treatment you need. Or you might also have other treatments before or after your surgery. This depends on the type of cancer you have, where it started, and if it has spread.

Curative surgery

Curative or primary surgery is usually done when cancer is found in only one part of the body, and all of the cancer probably can be removed. It is called **curative** because the purpose of the surgery is to remove all of the cancer and some healthy tissue around the cancer (the margin). Removing the healthy tissue increases the chance that all the cancer is removed. In this case, surgery can be the main or sometimes only treatment.

Surgery is often used with other treatments like [chemotherapy](#)⁴, [radiation therapy](#)⁵, [immunotherapy](#)⁶ or [targeted therapy](#)⁷.

- **Neoadjuvant** treatment is given **before** surgery to help shrink the tumor and make it easier to remove.
- **Adjuvant** treatment is given **after** surgery to treat any cancer cells that might remain in your body.

Debulking surgery

Debulking surgery removes part, but not all, of the cancer. Debulking removes as much of the cancer as possible. This surgery is done when all of the cancer cannot be removed.

Debulking surgery may be done if your cancer is too large or if removing all of the cancer would damage other parts of your body.

For example, it may be used for advanced [cancer of the ovary](#)⁸ and some [lymphomas](#)⁹. In these cases, the surgeon takes out as much of the cancer as possible and then treats what's left with radiation, chemotherapy, or other therapies. This might help decrease the amount of cancer so that other treatments can work better.

To relieve side effects (palliative or supportive)

This type of surgery is used to relieve side effects caused by cancer and to improve your quality of life. Examples are operations that can:

- Relieve pressure on nerves that are causing you pain
- Remove a block in your intestines or other area
- Prevent or fix broken bones
- Stop bleeding caused by your cancer

Palliative surgery helps ease problems caused by cancer and helps people feel better. This type of surgery is usually done when your cancer is at an advanced stage, so it's not done to cure the cancer itself.

Supportive surgery

Supportive surgery is done to make it easier for you to get other types of treatment. For example, a [vascular access device](#)¹⁰ such as an implanted port may be placed to give you treatments and draw blood instead of putting needles in your hands and arms each time you need IV fluids, blood transfusions, or treatments. An implanted port is a thin, flexible tube put into a large vein and connected to a small drum-like device that's placed just under your skin. A needle is put into the drum of the port when it needs to be used.

To repair or reconstruct body parts

Reconstructive surgery is used to improve the way a person looks after major cancer surgery. It's also used to restore the function of an organ or body part after surgery. Examples include [breast reconstruction after mastectomy](#)¹¹ or the use of tissue flaps, bone grafts, or prosthetic (metal or plastic) materials after [surgery for head and neck cancers](#)¹².

Reconstructive surgery may be done at the same time the cancer is removed or after you have healed or had other treatments.

To prevent cancer from developing

Preventive or prophylactic surgery can help lower your risk of getting some types of cancer by removing body tissue that's likely to become cancer, even if you have no signs of cancer.

Sometimes, an entire organ is removed when a person is at very high risk for having cancer in that area. For example, if you have a strong [family history](#)¹³ of breast cancer and have an inherited change in a breast cancer gene (called *BRCA1* or *BRCA2*), you might decide to have your breasts removed ([prophylactic mastectomy](#)¹⁴). Removing your breasts before cancer is found can help lower your risk of breast cancer in the future.

Preventive surgery is done to reduce your risk of getting cancer, but it doesn't guarantee cancer prevention. You will still need to have regular check-ups and [screening tests](#)¹⁵ to catch any cancer that does happen early.

[Watch this video on YouTube](#) ¹⁶

Types of cancer surgery

Many types of cancer are treated with surgery. There are many types of surgery, including those that require a large incision and others that use equipment so smaller incisions can be used. For some surgery, you won't have to have an incision at all.

The type of surgery you need will depend on your type of cancer, where it is in your body, and how big it is. You might need open surgery, which requires larger incisions (cuts). Or your surgeon might be able to remove your cancer using techniques that require smaller incisions, cause less pain, and have shorter recovery times.

Elective versus emergency surgery

Elective surgery is treatment that is planned ahead of time. Most cancer surgery is elective, meaning that you have time to get ready for the surgery.

Emergency surgery needs to happen quickly – often within hours or a couple of days. You might need emergency surgery if you develop a problem that can't be managed with medicines or other treatments.

Emergency surgery is done when problems are life-threatening. Some reasons for emergency surgery in people with cancer include:

- A bowel blockage (obstruction) so nothing can move through your intestines. This can cause vomiting, dehydration, and severe infection.
- A bowel tear or hole (perforation) that leaks bowel contents into your abdomen (belly), causing infection.
- A bone fracture that needs to be fixed so it doesn't move (stabilized) so it can heal.
- Bleeding, such as from the rectum, that other treatments can't manage.
- Swelling in the brain or spinal cord that must be relieved quickly.

Open surgery

Open surgery lets your surgeon look directly at the area of the body they think is affected by cancer. It also lets the surgeon see what other organs or lymph nodes are near the cancer. Open surgery can help your surgeon remove as much of your cancer as possible while decreasing the damage done to other organs.

Open surgery also allows your surgeon to get to parts of the body that less invasive procedures cannot.

Recovery from open surgery might take longer than less invasive types of surgery, and you might need help while you heal. Ask your surgeon or cancer care team what you can expect while you recover. Be sure to ask what kind of [support](#)¹⁷ you will need at home so you and your loved ones can make a caregiving plan. Having a plan can help reduce your stress while you recover.

Minimally invasive surgery

Your surgeon may think a less invasive surgical technique would be a good option for you. Minimally invasive surgery uses smaller incisions compared to open surgery. This approach also uses a camera, small instruments, and other equipment.

Minimally invasive surgery usually means that you will have a faster recovery time with less pain. You might still need someone to help care for you at home while you recover.

Laparoscopic surgery

During laparoscopic surgery, your surgeon makes several small cuts in the skin of your abdomen (belly). A thin, lighted tube with a camera is inserted through one of the cuts. Then, instruments are put through the other cuts to operate.

Laparoscopic surgery is sometimes used for biopsy procedures (taking pieces of tissue to check for cancer). It is also sometimes used to remove smaller cancers.

Using a [laparoscope](#)¹⁸ to remove some tumors can help reduce blood loss during surgery and pain afterward. It can also shorten your hospital stay and help you heal faster.

Laparoscopic surgery is now being used to remove some cancers, such as colon, rectum, cervix, uterus, ovaries, liver, prostate, and kidney. Studies are being done on uses in other types of cancer.

Thoracoscopic surgery

This type of surgery uses a [thoracoscope](#)¹⁹ to look inside your chest. It is a thin, lighted tube with a camera on the end that can be put through a small cut in your chest. The thoracoscope allows the surgeon to take biopsies of areas that might be cancer and drain fluid. This type of surgery can also be used to remove cancerous parts of your lungs. You may also hear this type of surgery called video-assisted thoracic surgery (VATS).

Thoracoscopic surgery requires less cutting than open surgery. You can usually recover more quickly with fewer problems than with open surgery.

Robotic surgery

A robotic system is sometimes used to help the surgeon during laparoscopic (or thoracoscopic) surgery. The surgeon sits at a control panel in the operating room with a monitor that shows the inside of your body. The surgeon then moves the robotic arms to operate through several small cuts in your body.

Robotic surgery is sometimes used to treat many of the same cancers that are treated with laparoscopic or thoracoscopic surgery.

Mohs surgery

Mohs micrographic surgery is used to remove certain skin cancers. One very thin layer is removed at a time. After each layer is removed, the tissue is checked for cancer cells with a microscope. This procedure is repeated until all the cells in a layer look normal.

Mohs surgery is used when it is not known how deep your skin cancer is or when as much healthy tissue as possible needs to be saved, such as when treating skin cancers on the face.

Procedures that don't need incisions

Laser surgery

A laser is a highly focused beam of light energy which can be used for very precise surgical work. It can be used instead of a scalpel (blade) to cut through tissue. It is most often used to treat cancers or precancers on the skin or the lining of internal organs that can be reached through normal body openings.

Laser surgery can help get rid of precancerous growths such as polyps in the colon or abnormal areas on the cervix. It may also be used to treat some cancers of the cervix, penis, vulva, esophagus, lung, and skin.

Lasers are also sometimes used to help relieve symptoms of cancer, such as when large tumors block the windpipe (trachea) or esophagus, causing problems breathing or eating.

Laser surgery can be faster than open surgery and less likely to damage surrounding tissues. You would usually heal faster and be less likely to have bleeding or swelling.

Learn more about the use of lasers in [Laser Therapy](#)²⁰.

Cryosurgery

Cryosurgery (cryotherapy) uses liquid nitrogen or a very cold probe to freeze and kill abnormal cells. This technique is sometimes used to treat precancerous conditions on a body surface, like those on the skin, cervix, and penis.

Cryosurgery can also be used to treat some deeper cancers, like those in the liver and prostate. A scan (like an [ultrasound](#)²¹ or [CT scan](#)²²) might be used to guide the cryoprobe to where the cancer cells are. This limits damage to nearby healthy tissue.

Radiofrequency ablation

Radiofrequency ablation, or RFA, is a type of [hyperthermia](#)²³ - a treatment that uses heat to destroy cancer cells. In RFA, high-energy radio waves are sent through a needle to heat and destroy cancer cells. RFA may be used to treat cancer tumors in the liver, lungs, kidney, and other organs.

Stereotactic radiation therapy

Also called **stereotactic radiosurgery**, stereotactic radiation therapy is not surgery but a way to give radiation therapy. It's sometimes called surgery because it's so precise, even though no cut is actually made.

You can learn more about this kind of treatment in [Radiation Therapy](#)²⁴

What is it like to have surgery?

[Getting Ready for Cancer Surgery](#)

Preparing for cancer surgery can be overwhelming. Learn about steps taken to prepare for surgery.

Risks and Side Effects of Cancer Surgery

Before you decide to have surgery for cancer, it's important to understand the risks. Learn about those risks & possible side effects here.

Anesthesia for Cancer Surgery

Learn more about the types of anesthesia, when each is used and what recovery is like.

Recovering from Cancer Surgery

Recovering from cancer surgery can be overwhelming. Learn what you might expect while recovering.

Choose a cancer type Acute Lymphocytic Leukemia (ALL) Acute Myeloid Leukemia (AML) Adrenal Cancer Anal Cancer Basal and Squamous Cell Skin Cancer Bile Duct Cancer Bladder Cancer Bone Cancer Brain and Spinal Cord Tumors Brain and Spinal Cord Tumors in Children Breast Cancer Cancer of Unknown Primary Cervical Cancer Childhood Non-Hodgkin Lymphoma Chronic Lymphocytic Leukemia (CLL) Chronic Myeloid Leukemia (CML) Chronic Myelomonocytic Leukemia (CMML) Colon and Rectal Cancer Endometrial Cancer Esophageal Cancer Ewing Tumors Eye Cancer Gallbladder Cancer Gastrointestinal Carcinoid Tumors Gastrointestinal Carcinoid Tumors Gastrointestinal Stromal Tumor (GIST) Kidney Cancer Laryngeal and Hypopharyngeal Cancer Leukemia in Children Liver Cancer Lung Carcinoid Tumor Melanoma Merkel Cell Skin Cancer Mesothelioma Multiple Myeloma Nasal Cavity and Paranasal Sinus Cancer Nasopharyngeal Cancer Neuroblastoma Non-Hodgkin Lymphoma Non-small Cell Lung Cancer Oral Cavity and Oropharyngeal Cancer Osteosarcoma Ovarian Cancer Pancreatic Cancer Pancreatic Neuroendocrine Tumor (NET) Penile Cancer Pituitary Tumors Prostate Cancer Retinoblastoma Rhabdomyosarcoma Salivary Gland Cancer Small Cell Lung Cancer Small Intestine Cancer Soft Tissue Sarcoma Stomach Cancer Testicular Cancer Thymus Cancer Thyroid Cancer Uterine Sarcoma Vaginal Cancer Vulvar Cancer Wilms Tumor

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Getting Ready for Cancer Surgery

It is normal to feel nervous or worried as you get closer to having surgery. Knowing what to expect before, during, and after your surgery can help you prepare and feel more at ease.

Your surgery experience will depend on many things, including the type of cancer you have, the type of surgery you are getting, and your overall health. Knowing what to expect and being prepared can help. It's important to:

- Learn as much as you can beforehand
- Ask questions so you know what to expect
- Know that each person's situation is different

How quickly will I need to have surgery?

Where will the surgery be done?

Who will do my surgery?

Getting ready for surgery

What happens during cancer surgery?

How quickly will I need to have surgery?

If you need to have surgery for your cancer, you might wonder if you need to get it right away or if you can wait a bit.

How soon you need surgery after finding out you have cancer depends on your type of cancer and other factors. Sometimes, cancer surgery needs to happen as soon as possible. Other times, waiting a while is not a problem. And sometimes, you might need [chemotherapy](#)¹ or [radiation](#)² before having surgery.

Talk to your surgeon or others on your cancer care team about how quickly you need to have surgery. Don't be afraid to ask questions. You might want to ask if you have time to think about other options or get a [second opinion](#)³.

Where will the surgery be done?

Where your surgery is done depends on the type of surgery you need, and how much time you will need to recover under the care of your surgical team. Surgery can be done in a doctor's office, clinic, surgery center, or hospital.

Outpatient or ambulatory surgery: If you don't need to be monitored for a long time after your surgery, you can go home the same day or the next day.

Inpatient surgery: If you need to be watched by the surgical care team for more than one day, you might need to stay at the hospital.

Your surgeon will let you know where your surgery will be done and how long you'll need to recover after the procedure. You will also know if you'll need to stay in the hospital after surgery and for how long.

Who will do my surgery?

The type of surgeon who does your operation depends on the [type of cancer](#)⁴ you have and where it is in your body. Your choice of surgeon might also depend on the hospital or treatment center where you get your surgery.

Surgical oncologists perform many types of cancer surgery. Some may specialize in certain types of cancers, such as breast or colorectal, but can often treat other types of cancer.

Some specialize in treating cancer in specific parts of the body such as:

- Breast surgeons focus on treating people with breast cancer.
- Dermatologists or dermatologic surgeons mainly treat people with skin cancer and other skin problems.
- Gastrointestinal (GI) surgeons treat cancers of the stomach, small intestine, large intestine (colon), and rectum. They may also treat other cancers in the abdomen (belly), such as pancreatic, liver, and gallbladder.
- Gynecologic oncologists treat cancers of the female reproductive system, such as cervical cancer, ovarian cancer, and uterine cancer. One common surgery done by this type of surgeon is a [hysterectomy](#)⁵ which removes the uterus and part of the cervix.
- Neurosurgeons are specially trained to treat cancers in the brain or spine.
- Oral and maxillofacial surgeons treat people with cancers affecting their mouth, jaw, or face.

- Orthopedic surgeons focus on treating different types of cancer in the bones or repairing damage to the bones from cancer.
- Otolaryngology or ear, nose, and throat (ENT) surgeons treat people with cancers of the ear, nose, and throat.
- Plastic or reconstructive surgeons specialize in changing how a body part looks after cancer treatment. They may rebuild or replace removed or injured body parts.
- Thoracic surgeons focus on surgery for cancers in the chest, such as the lungs or esophagus.
- Urologists and urologic oncologists focus on treating cancer of the urinary tract. This includes bladder, kidney, prostate, and testicular cancers.

When you are choosing the doctor who will do your surgery, be sure to ask how much experience they have in treating your kind of cancer. You can ask for a second opinion if you wish to talk to another surgeon before deciding.

Who will take care of me during surgery?

Your surgeon will lead the team who take care of you before, during and after your surgery. You might not meet some of them, but they will be in the operating room during your surgery or in the recovery room after your surgery is finished. Besides your surgeon, other members of your surgical team might include:

Anesthesiologist. This type of doctor cares for patients right before, during, and after surgery by giving [anesthesia](#) and monitoring their vital signs, such as breathing, heart rate, and blood pressure.

Certified registered nurse anesthetist (CRNA). This specially trained nurse might give you anesthesia and monitor your vital signs. They might work with the anesthesiologist or on their own.

Oncology nurse practitioners (NPs) and oncology physician assistants (PAs). These providers work alongside a surgical oncologist. They might:

- Do physical examinations before and after surgery
- Order and interpret laboratory and diagnostic test results
- Prescribe medicines
- Provide education to you, your caregiver, and loved ones
- Assist in surgery and with other procedures

Operating room nurses. These nurses may assist the surgeon during surgery.

Surgical technologists. These professionals might assist the surgeon during surgery. They often get the operating room ready and help get you ready for surgery.

Recovery room nurses and staff. These nurses and other staff care for and monitor people who have just had surgery as they recover.

Other health care professionals. Other team members such as physical therapists, pharmacists, social workers, and nutritionists may help you recover and get the care you need before and after surgery.

Meeting your surgical team

When you meet with your surgeon or surgical team, they will:

- Review your medical record
- Examine you
- Evaluate the need for surgery
- Order some tests to see if you're healthy enough for surgery
- Explain the risks, benefits, and other options (if any) than the surgery
- Consult with other specialists to collaborate in your care (e.g., plastic and reconstructive surgeon, medical oncologist, radiation oncologist, etc.)
- Give you hints, tips, and pointers to get you organized and ready for surgery and the recovery period
- Give instructions for your care after returning home, including what you will and won't be able to do
- Answer any questions you have

Getting ready for surgery

The time before surgery is called the *pre-operative phase*. Although there are many kinds of surgical procedures, certain steps need to happen during the pre-operative phase for almost all of them. Your surgical team will manage this part of your care.

Insurance pre-approval

Before agreeing to surgery, ask your surgeon or other member of your cancer care team whether you will need pre-approval (or prior authorization) for the surgery. They might suggest that you check with your insurance provider to see if this is needed.

The facility where you are getting surgery may have a billing department that can help you figure this out and prepare any needed paperwork. Your surgeon or other members of the surgical team can tell you if there is someone who can help you with this.

Give your permission (informed consent)

Before your surgery can be done, you and your surgeon must go through a process called [informed consent](#)⁶. This means that your surgeon and other members of the surgical team must talk to you about what you can expect with your surgery and answer any questions you have. Your surgical team should talk to you about:

- Your treatment options, including surgery and other treatment types
- How long it will be before surgery can be done
- The operation, including the benefits, risks, and side effects
- What to expect before, during, and after surgery

Once you have gotten this information and had your questions answered, your surgeon will ask if you understand what to expect and agree to the surgery. You will then be asked to sign a consent form. Signing this form means:

- You give written permission for this surgery
- Your team gave you information on your surgery, its risks, and other treatment options
- All your questions have been fully answered (for the time being)
- You choose to have this surgery
- You understand that the surgeon can't guarantee that the surgery will give the expected results

Other factors that could affect your surgery

Tobacco: If you smoke, your surgeon may ask you to stop before surgery. Using tobacco tightens (constricts) blood vessels and decreases the amount of oxygen that gets to your body tissues. Smoking can delay healing and recovery. It can also increase the risk of complications after surgery.

Diet and alcohol: Excess weight or obesity may affect surgery and recovery. Your surgeon may ask you to improve your diet, lose weight, or exercise before surgery. You may be advised to stop drinking alcohol, too.

Ask your surgery team about any diet restrictions before surgery. You may be told not to eat or drink the day or days before your operation or you might need to avoid certain foods or fluids. Restrictions are common, especially 24 hours before surgery.

Physical activity: Being physically active can also help you build your strength and gain more energy before you have surgery. It can also help you deal better with stress and anxiety that surgery can cause. Being more active before surgery might help you recover more easily and have fewer complications.

Medicines and drugs: Be sure to tell your surgeon and surgical team about any prescriptions, over-the-counter medicines, vitamins, supplements, and marijuana or street drugs you may take. Some of these might cause problems before, during, and after surgery.

You may need to stop taking certain medicines before surgery. Some medicines must be stopped a couple of weeks or days before surgery. Others can be stopped the day before surgery, or can be taken as usual. Often, the surgeon will ask you to stop taking certain medicines, such as anti-inflammatory pain medications and blood thinners. Those medicines can increase your risk of bleeding during the surgery.

Anesthesia history: You will probably be asked if you or any family members have had problems with anesthesia before. There are things that can be done to prevent certain problems, such as nausea, vomiting, and being overly sleepy after getting anesthesia.

Pre-operative testing

You'll have tests to check your overall health and see if you have any problems that need to be watched for during surgery. The tests you need will depend on your situation, but here are some standard tests that might be done:

- Blood tests to check your blood count, blood sugar, kidney and liver function, and your risk for bleeding
- Blood test to identify your blood type in case you need a blood transfusion
- Urine test to make sure your kidneys are working and to check for infection
- Chest x-ray to look at your lungs
- Electrocardiogram (ECG or EKG) to test your heart function

To help plan your surgery, you may need imaging tests, such as magnetic resonance imaging (MRI), computerized tomography (CT) scan, positron-emission tomography (PET) scan, an ultrasound, or a bone scan.

Will I have to stay in the hospital after surgery?

Depending on the type of surgery and how well you do, you may be able to go home the same day or have to stay in the hospital overnight.

Outpatient surgery means you can go home the same day as your surgery. You will need someone to take you home so plan to bring a family member or friend. Your surgeon or surgical team will follow-up with them after surgery to let them know how the surgery went and what you will need when you get home.

Inpatient surgery means you must stay in the hospital for at least one night after surgery. Many people like to bring a family member or friend to wait during surgery. Your surgeon or surgical team will talk with them during or after surgery to let them know how it went.

Your surgical team will let you know what kind of aftercare and support you will need at home after surgery. Ask your surgical team when you should see the surgeon for a follow-up visit. **Also ask how and who to call, in case you have a problem, or a question comes up after the surgery.** Be sure to ask your team about these things before leaving the place you had surgery.

Prep before surgery

A "prep" may be needed before surgery if you are getting anesthesia. You will likely be told to stop eating and drinking for a certain time before surgery. Sometimes, you will be told when to stop eating solid foods, and then liquids will be stopped later.

For some surgery, you might have to take a laxative or enema to be sure your bowels are empty. Your skin will need to be cleaned well before the operation to reduce the risk of infection. You may be asked to shower or bathe the day before or morning of your surgery. Some people will be told to use a specific cleanser so check with your surgical team.

What should I wear to my surgery? What should I bring with me?

Your surgical team will let you know what to wear and bring with you to your surgery. You may be asked to remove and store all clothing and/or jewelry, including rings, before surgery. Think about leaving jewelry and other valuables at home or with a family member or friend in the waiting area.

If you wear contacts, be sure to wear eyeglasses on the morning of your surgery. It is easier to remove your glasses. You can bring your contact lenses, case, and solution to use after surgery.

You might be asked not to wear makeup, skin lotions, or creams on the day of your surgery. This will depend on the type of surgery you are having, so be sure to check your instructions and ask questions about anything unclear.

What happens during cancer surgery?

The day of your surgery, the surgical team will get you ready for surgery. Some of the surgical preparation happens before you receive anesthesia and some happens after. People having surgery often receive anesthesia. There are different types of anesthesia depending on the type of surgery.

Before surgery, the staff will help you change into a surgical gown and you may be given a cap to wear on your head and socks. The staff may also remove any hair from the surgical area and clean your skin. This helps reduce the risk of [infection](#)⁷, and is an important part of care after surgery.

You may also have a catheter placed. A catheter is a flexible tube that is inserted into the body to collect urine from the bladder.

During many surgical procedures, the surgeon will remove your cancer. The surgeon often also removes some healthy tissue around your cancer to improve the chances that all the cancer is removed. There are many types of cancer surgery, depending on where your cancer is and how big it is. For some types of cancer surgery, nearby lymph nodes may also be removed.

During surgery, your doctor may insert one or more drains to help keep fluid from building up at the incision site. These drains may be removed before surgery is complete or may be left in place after the procedure is completed.

If you will still have the drain when you leave the hospital or clinic, your surgical team will tell you how to care for it. Be sure to ask any questions about your drain or your care after surgery.

Once the cancer is removed and drains (if used) are placed, your doctor will close your incision with stitches, staples, surgical glue, or special bandages. A dressing is often then placed to cover your incision. Your surgical team will teach you how to care for your incision when you go home.

Hyperlinks

1. www.cancer.org/cancer/managing-cancer/treatment-types/chemotherapy.html
2. www.cancer.org/cancer/managing-cancer/treatment-types/radiation.html
3. www.cancer.org/cancer/managing-cancer/making-treatment-decisions/seeking-a-second-opinion.html
4. www.cancer.org/cancer/types.html
5. www.cancer.org/cancer/types/endometrial-cancer/treating/surgery.html
6. www.cancer.org/cancer/managing-cancer/making-treatment-decisions/informed-consent.html
7. www.cancer.org/cancer/managing-cancer/side-effects/infections.html

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Anesthesia for Cancer Surgery

Anesthesia is the use of medicines to cause a loss of feeling or sensation during surgery or other procedures. This loss of feeling might only be in the surgical area, or you might lose consciousness (fall asleep).

Learn more about the types of anesthesia, when each is used and what recovery is like.

[Local anesthesia](#)

[Topical anesthesia](#)

[Regional anesthesia](#)

[Sedation](#)

[General anesthesia](#)

Depending on the type of surgery you are going to have, your surgeon will suggest a type of anesthesia. Sometimes, you may get to choose the type of anesthesia used.

Local anesthesia

Local anesthesia numbs only a certain area, so you won't feel pain there. It is often used for minor surgeries, such as biopsies of areas near the surface of your body.

A needle is used to inject medicine into the area. This numbs the nerves that cause pain. You stay awake and usually feel only pressure during the procedure. You can usually go home soon afterward.

Topical anesthesia

Topical anesthesia can cause numbness when rubbed or sprayed onto a body surface instead of being put in with a needle. For example, a spray is sometimes used to numb your throat before a scope is passed down to the stomach or lungs. Same as with local anesthesia, you can usually go home soon after the procedure.

Regional anesthesia

Regional anesthesia (such as a nerve block or spinal anesthesia) numbs a larger part of the body, but you stay awake. For example, a needle can be used to put medicine into an area around the spinal cord to affect certain nerves coming out of it.

A nerve block is an injection of medicine around the nerves in the arms or legs. Where the injection is given is based on the part of the body that needs to be numbed.

Regional anesthesia might be given as a single injection or an ongoing IV infusion. You stay awake, but you may be given something to help you relax. You will go to the recovery room after your surgery until the anesthesia wears off.

Sedation

Sedation medicine make you feel relaxed and sleepy. It may be given a pill or liquid by mouth or through an IV. Sedation might be given alone or along with local or regional anesthesia. It can make you more comfortable and reduce pain.

There are 3 levels of sedation:

Minimal sedation makes you relaxed. If you feel anxious or nervous, it helps you feel comfortable during a test or procedure. This level of sedation means you are awake during the procedure and can answer questions.

Moderate sedation reduces pain and makes you less aware of things around you. This is also called **conscious** or **twilight** sedation. You might be able to talk and respond to questions, or you might doze off.

Moderate sedation might be used for minor surgery, along with local or regional anesthetic. You might not remember the surgery and the time right after. You will be in the recovery room after surgery until some of the anesthesia wears off.

Deep sedation makes you unaware of the procedure. This usually means you can't talk or answer questions. You may also receive pain medication to reduce any discomfort.

The medicines used for deep sedation might affect your memory so you don't remember the surgery. You might be given oxygen or need a breathing tube to help you breath during deep sedation. You will go to the recovery room after surgery until some of the anesthesia wears off.

General anesthesia

General anesthesia puts you into a deep sleep so you are unconscious for the surgery. Because of the medicines given, you won't remember the surgery and the time right after is often very hazy.

What happens when you get general anesthesia?

You will be given general anesthesia in these ways:

- Through an IV line placed in a vein in your arm
- Through a face mask or tube in your windpipe, so you will breathe in the medicine
- Both of these methods

Anesthesia often begins to take effect in less than 1 minute. Once you are asleep, an endotracheal or ET tube is put in your throat to help you breathe during surgery.

Your heart rate, breathing rate, and blood pressure will be watched closely during the surgery. A doctor or nurse who specializes in giving anesthesia (either an anesthesiologist or nurse anesthetist) takes care of you while you are asleep. They will also keep track of your anesthesia level to make sure you do not wake up until the time is right.

The anesthesiologist will stop the anesthesia and take out the ET tube when your procedure is over.

You might also be given medicine to help you wake up. Next, you will go to a recovery area (post-anesthesia care unit – PACU) until some of the anesthesia wears off. Specially trained nurses will take care of you as you finish waking up.

What are the possible side effects of general anesthesia?

Anesthesia is safe for most people. But as with any medicine, you might have side effects. They can include:

- Feeling groggy, sleepy, or confused
- Nausea or vomiting
- Shaking chills

- Itching
- Blurry vision
- Dry mouth or sore throat, especially if you needed a tube to help you breathe during the procedure
- Trouble emptying your bladder

These side effects usually go away on their own in the PACU. Your surgical team will help you manage them. The nurse in the PACU is there to watch how you are waking up and will decide if and when to offer you something to drink.

Serious complications of anesthesia are rare, but they include serious injury and death. The risks can be higher for older adults, especially people with severe heart or lung disease.

Another rare side effect is anesthesia awareness. This means a person briefly wakes up during the procedure or remembers what happened at times during surgery. This happens in no more than 2 out of every 1,000 people. Risk factors for anesthesia awareness are:

- Having emergency surgery
- Use of certain types of anesthesia
- Using certain medicines for a long time, including tranquilizers, opioids, and medicines called anti-convulsants
- Having excess weight or obesity
- Daily alcohol use

What is recovery from general anesthesia like?

It could take you some time to feel like yourself after general anesthesia. Your surgical team will let you know how long you need to be monitored after your surgery. This will depend on the type of surgery you had and your overall health.

Your reaction speed and judgment may be slow or different for a while after anesthesia. Avoid these activities for at least 24 hours after waking up from surgery:

- Driving a car
- Operating other heavy machinery
- Making important decisions

Ask your health care team when you can start taking your regular medicines again. Also ask if you need to be careful about what you should eat and drink. For instance, you may have trouble digesting some foods. Start by drinking liquids and eating light food until you feel normal again. Examples of light foods include:

- Broth or soup
- Tea or black coffee
- Yogurt
- Gelatin
- Toast, crackers, or plain rice

If you feel you are having trouble recovering from anesthesia, contact your surgical team right away.

Who will give the anesthesia?

Who gives you anesthesia depends on the type of anesthesia you will receive.

Anesthesiologist. An anesthesiologist is a doctor who specializes in giving anesthesia and caring for people while they are getting it. If you are receiving certain types of anesthesia, such as general anesthesia, an anesthesiologist will lead a team of professionals who will monitor you while you are having the procedure.

Certified registered nurse anesthetist (CRNA). This specially trained nurse might give your anesthesia and monitors your vital signs. They might work with the anesthesiologist or on their own.

Surgeon or other doctor. In some cases, the doctor performing the surgery may administer the anesthesia themselves. This is especially true if it is topical or local anesthesia.

Anesthesia assistant. The assistant has a college degree and additional training in anesthesia care. They specialize in taking care of patients during and after anesthesia.

How to get ready for anesthesia

Your anesthesiologist may meet with you before you have general anesthesia and ask you some questions. They might meet with you before or on the day of your surgery. They will likely ask you:

- About your health, now and in the past
- What medicines, vitamins, and herbal supplements you take, including prescription and over-the-counter products
- If you have allergies, including to foods, medicines, and items like latex
- If you have had anesthesia before, and your experience with it
- If you use other drugs, alcohol, or tobacco
- If you have any loose teeth, dentures, or dental bridges. Some types of anesthesia mean that you might need help breathing and your mouth may be moved around. Your team will want to protect your teeth.

This information helps your surgical team give you anesthesia as safely as possible.

It is important to follow your surgical and anesthesia teams' instructions. You will probably need to not eat anything for at least 6 hours before general anesthesia. You may be able to drink clear liquids until a certain time. Do not eat or drink **anything** unless your doctor tells you it is OK.

You might need to stop taking some of your regular medicines before general anesthesia. Some medicines can affect the anesthesia or your procedure. Ask your doctor which medicines you should take and which ones you should not take. Also, make sure you follow their instructions for starting the medicine again.

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Risks and Side Effects of Cancer Surgery

Any medical procedure has risks. Before you decide to have surgery or any other procedure, it's important to understand the risks and what will be done to reduce them.

Each procedure has its own risks and side effects. Be sure to ask your surgical team about your risks. The expected benefits of the surgery should outweigh the possible risks.

[Safety before and during surgery](#)

[Possible side effects of cancer surgery](#)

[Dealing with changes to your body from cancer surgery](#)

Possible long-term side effects of cancer surgery

Can surgery cause cancer to spread?

Your surgical team will take many steps to reduce your risk of side effects and complications. Special leg pumps (pneumatic or compression stockings) and low-dose blood thinners might be used to avoid blood clots, and breathing treatments (respiratory therapy) to help prevent pneumonia. Ask your doctor about the possible complications of your surgery and what will be done to help prevent them.

Safety before and during surgery

There are always risks if you are having surgery. Your surgeon and surgical team will talk to you about these before you give your permission for the surgery.

Hospitals and surgical centers have protocols to reduce some of these risks such as surgical checklists and standard steps for communication. These are for your safety even if they seem repetitive. Some of the steps to prevent common risks include:

- Making sure that you are the person who should be getting the surgery. The surgical and operating room staff will ask you to identify yourself by stating your name and birthdate, which is compared to the ID band you got when you checked in or were admitted.
- Making sure that you get the right surgery. You will be asked to confirm that you are supposed to be getting the surgery planned. Your surgeon may also mark the site for the surgery on your body with a marker. This is most important if your surgery is on a specific side of your body, such as a mastectomy on your right breast.
- Taking steps to lower the risk of infection after your surgery. Your skin will be cleansed with special products to get rid of any pathogens that could cause infection. You might also be given antibiotics before, during, and sometimes after surgery.
- Monitoring your body temperature and keeping it in a safe range. This can help prevent some side effects of surgery that might make recovery harder.
- Performing checks to make sure no surgical equipment is left in your body. Operating room staff do equipment counts before your surgery is finished.

If you have questions or are worried, be sure to ask your surgical team before your operation.

Possible side effects of cancer surgery

Side effects from surgery can be caused by the surgery itself, the medicines used, and your overall health. In general, the more complex the surgery, the greater the risk of side effects.

Minor operations and taking tissue samples (biopsies) usually have less risk than a bigger surgery. Pain at the surgery site is the most common problem. Infections at the site and reactions to the medicines used to numb the area (local anesthesia) are also possible.

Side effects are possible during and after surgery. These side effects usually are not life-threatening and get better with treatment.

Appetite loss

You might not feel like eating and drinking as much as you usually do. Poor appetite is common, especially if you had general anesthesia. You may lose weight.

Your appetite almost always returns and you will be back to your normal weight as the effects of surgery wear off. If your appetite does not come back, tell a member of your surgical team.

Bleeding

Bleeding can happen with any surgery and is usually controlled. Bleeding can happen either inside the body (internally) or outside the body (externally). Bleeding can occur if a blood vessel was not sealed off during surgery or if a wound opens up.

You may have some bleeding from your incision after surgery. Your surgical team will tell you how to take care of your incision and when to let them know if you are bleeding. Your wound will likely be covered with a dry bandage.

If you are bleeding a lot, put pressure on your incision site until you can get to your surgeon's office or the emergency room. It may help to put more gauze over your bandage to soak up any extra blood.

Your surgical team will try to limit bleeding when working near blood vessels. They may also check lab tests to see if your blood is less likely to clot normally.

It does not happen often, but if your bleeding is severe, you might need to have another operation to find and stop its source. If you bleed enough, you might need a blood transfusion to replace what you've lost.

Blood clots

You can get blood clots in the deep veins of your legs after surgery. This is why it's important to get out of bed and sit, stand, or walk as soon as possible if permitted by your surgical team.

Blood clots that occur in veins are called a **venous thromboembolism (VTE)**. If the vein is in the leg, thigh, or pelvis, it is called **deep venous thrombosis (DVT)**. A clot like this can become a serious problem if it breaks loose and travels to another part of the body, such as a lung (pulmonary embolism).

You may be given medicines called **anticoagulants** to help prevent a blood clot after surgery. You may also be asked to wear compression devices or stockings on your lower legs to help prevent blood clots until you can walk around.

If you develop [symptoms of a blood clot](#)¹, it is important to get medical help right away. Let your surgeon know or go to an emergency department if you have:

- Arm or leg swelling on only 1 side of the body
- Pain in the arm or leg where a blood clot is located
- Trouble breathing or chest pain when breathing
- Rapid heartbeat

Bruising

You may have bruising around the surgery site. After an incision, some blood can leak from small blood vessels under the skin. Let your surgeon know if you have swelling along with the bruising.

Drainage

You may have some drainage from the incision. Sometimes, the fluid that builds up at the surgery site drains through the surgical wound. These symptoms can be signs of an infection. Tell someone on your surgical team right away if you have any of these symptoms:

- Drainage that smells bad
- Redness around the incision
- Fever

Fatigue

Fatigue is physical and mental tiredness or exhaustion. You might have fatigue because of the anesthesia, the energy needed to heal, loss of appetite, and stress.

Fatigue usually goes away over 2 to 4 weeks after surgery. Ask your surgical team what is typical for the type of surgery you will have and ways to cope with fatigue.

Infections

An [infection](#)² might develop at the site of the incision, but it can also happen in other parts of your body.

You may be asked to help prevent infection by washing with a special soap for a few days before surgery. This soap is good at killing bacteria and can help prepare your skin for surgery. Your surgical team will also try to prevent infection at the site of the incision (cut). You may be given antibiotics, either as a pill or through a vein in your arm (IV), to help prevent infection.

You are also at risk for other infections after surgery. You might get a lung infection (pneumonia), especially if you have a chronic lung illness or if you smoke. Doing deep breathing exercises as soon as possible after surgery helps lessen this risk.

Other infections can develop in your body, especially if your stomach or intestines were opened during the operation. Having a catheter to drain urine can increase the risk of a bladder infection, especially if it's left in place for a while.

Your surgical team will check for infection and monitor any changes in your temperature, skin, or incision to try to catch and treat any infection early. Signs of infection in a surgical incision include:

- Redness
- Warmth
- More pain
- Drainage from the wound
- Fever

If you have any of these signs, tell someone on your surgical team right away. You may be given antibiotics to take by mouth or antibiotic cream to treat signs of an infection. Antibiotics generally work well to treat most infections.

Occasionally, an infection can become an abscess. An abscess is a closed cavity filled with fluid or pus. A doctor usually has to drain an abscess. Antibiotics alone do not work well for an abscess because the medicine may not be able to reach the infection.

Numbness

Because nerves in the skin are cut during surgery, you may have some numbness. This doesn't usually cause any problems, but it can last a long time or sometimes even become permanent. Ask if any nerves will be affected by your surgery and, if so, for how long.

Pain

Almost everyone has some pain after surgery. How much pain you have and where it's located depends on where on your body the surgery was done, the size of the cut or incision, and the amount of tissue removed.

Pain after surgery is normal, but it should not be allowed to slow down your recovery. There are many ways to help control surgical pain. Medicines for pain can range from acetaminophen to anti-inflammatory medicines or stronger medicines, like morphine.

See [Pain³](#) for more information.

After surgery, pain usually goes away gradually as your body heals. Let your health care team know how you are feeling as you recover.

Reactions to medicines

Some people react badly to the anesthesia or other medicines needed during surgery. Although rare, these reactions can be serious because they can cause low blood pressure. Your heart rate, breathing rate, blood pressure, and other signs will be watched closely during surgery to prevent, look for, or correct this.

Slow body function recovery

Some of your body functions, such as bowel activity, can be slow to recover after surgery. If it takes too long for these functions to recover, serious problems might develop. Your energy level can drop, too. Getting out of bed and walking around as soon as possible after surgery can help you recover more quickly.

Surgery on certain parts of the body, such as the abdomen or chest, may cause short-term problems with surrounding organs. These problems are rare but can be serious. They are more likely to happen to people who already have problems with these organs. This is why doctors get a complete medical history and order tests to look for possible risks before surgery.

For example, the intestines may become paralyzed for a short time after surgery in the abdomen (belly). This means that the intestine will not allow food, fluid, and gas to pass through the bowels. It can cause nausea and vomiting, stomach cramps, and bloating until the bowels begin to function again.

These problems usually go away as you heal. Let your surgical team know if you are still having these or other problems. They can help you manage these symptoms.

Swelling

It is normal to have some swelling around your incision. An incision is a form of injury and the body responds to injury with inflammation. As you heal, swelling usually goes away.

Dealing with changes to your body from cancer surgery

Cancer surgery may change how your body looks, feels, and functions. This can affect your body image, or how you feel about your body. Some people might feel upset or insecure and struggle with the changes to their self-image. If a part of your body does not look like you expected it to after surgery, you might feel sad or self-conscious.

It is as important to treat the [emotional side effects](#)⁴ of cancer surgery as the physical side effects. Here are some tips to help you prepare for and cope with body changes that affect how you feel about yourself.

Before your surgery

- Gather information about how surgery will affect your body, appearance, and physical abilities. Ask your surgical team for details on what to expect. This can give you time to adjust to the changes.
- Ask about options for [reconstructive surgery](#) or [prostheses](#)⁵ if you need to have a body part removed or altered. A prosthesis is an artificial body part. Ask your surgical team for details on what to expect.
- Ask if there are situations that might come up during your surgery that could cause a different outcome.
- Ask about your options for [cancer rehabilitation](#)⁶ and other ways to improve your recovery from surgery.

After your cancer surgery:

- Follow your surgical team's instructions for post-surgery care, including wound care, nutrition recommendations, and physical activity.
- Give yourself the time to heal and take part in activities that you enjoy, such as journaling or creating art.
- Let your surgical or cancer care team know how you feel, both physically and emotionally.
- Consider talking with a [counselor](#)⁷ who can help you cope with changes to your body.
- Join a [support group](#)⁸.

The changes to how your body looks and works after surgery will often depend on your type of cancer. Learn more about how your surgery for your type of cancer will affect your body.

Possible long-term side effects of cancer surgery

Ask if there could be any long-term or permanent effects from the surgery. Long-term side effects depend on the type of surgery done. You might want to ask if your surgery will:

- Affect your [fertility](#)⁹ (your ability to have a baby or father a child)
- Affect your ability to have sex like you're used to.
- Leave you with any sort of [ostomy](#) (opening through which stool or urine drain, or through which you will breathe).
- Cause problems controlling your urine or emptying your bladder.
- Increase your risk for [swelling](#)¹⁰ in your arms or legs.
- Numbness

Your surgeon should talk to you about the possible long-term effects of surgery before your operation.

Can surgery cause cancer to spread?

The chances that surgery would cause your cancer to spread are very low. Surgeons take special care to decrease this risk. Still, this could happen in some situations. Doctors who have experience in treating cancer with surgery are very careful to avoid these situations.

In the past, larger needles were used to take a piece of the tumor (biopsy) to diagnose cancer. The chance of spread or “seeding” from the biopsy was higher. Now, it’s more likely that a small needle will be used to biopsy your tumor. This, and other improvements to the technique, have greatly lowered the chances of a biopsy causing a cancer to spread.

Sometimes an incisional biopsy is used to get tissue. This is where the surgeon cuts through the skin to remove a small part of the tumor. But this may not be safe for a few types of cancer such as certain tumors in the eyes or in the testicles. Doctors may treat these types of cancer first, without taking a biopsy, or may recommend removing (resecting) the entire tumor if it’s likely to be cancer.

There are a few kinds of tumors that have a bit higher risk of cancer spread when they are cut out. Examples include parathyroid and gallbladder tumors, and some sarcomas. However, this is rare because of advances in equipment and imaging tests.

Exposure to air during surgery

A common myth is that cancer will spread if it’s exposed to air during surgery. This myth may be because people feel worse after surgery than they did before. But it’s normal to feel this way when recovering from any surgery.

People may also believe that air can cause cancer to spread because more cancer may be found during surgery than was seen on scans and x-rays. This isn’t because of the surgery – the cancer was already there; it just didn’t show up on the tests.

Cancer does not spread because it has been exposed to air. If you delay or refuse surgery because of this myth, you could be harming yourself by not getting effective treatment before the cancer spreads on its own.

Hyperlinks

1. www.cancer.org/cancer/managing-cancer/side-effects/low-blood-counts/blood-clots.html
2. www.cancer.org/cancer/managing-cancer/side-effects/infections.html
3. www.cancer.org/cancer/managing-cancer/side-effects/pain.html
4. www.cancer.org/cancer/managing-cancer/side-effects/sexual-side-effects/pelvic-surgery.html
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Recovering from Cancer Surgery

Right after your surgery, you will be moved to the recovery area. How long you will stay in the recovery room depends on the type of surgery you had and the type of anesthesia used.

[After the anesthesia wears off](#)

[Going home after surgery](#)

[When to call your doctor after cancer surgery](#)

After the anesthesia wears off

- After **local** anesthesia, you may be able to go home shortly after the procedure.
- After **regional** anesthesia with conscious sedation (which reduces pain, relaxes you and may make you sleepy), your team will monitor you carefully until the anesthesia wears off. How long it takes depends on the extent of your surgery.

- After **general** anesthesia, you will be monitored carefully in a recovery room until the anesthesia wears off. This usually takes 1 to 2 hours, but could take longer depending on the type and extent of your surgery. As anesthesia wears off, you may feel groggy, have a sore throat, or have pain or soreness at the site of your surgery.

Once you recover from anesthesia, you will be taken to the ICU, your hospital room, or you might be allowed to go home. Make sure you follow all recovery instructions from your surgical team.

Let your surgical and cancer care teams know how your recovery is going and let them know if you have problems or concerns. This way, they can help relieve or avoid the side effects of surgery.

Tubes and catheters

Your throat may be sore for a while if you had an endotracheal (ET) tube to help you breathe after surgery. You might also have tube (called a **Foley catheter**) draining urine from your bladder into a bag. This is usually taken out as soon as possible after surgery to prevent infection.

Incision site care

You will likely have a bandage over your surgical site. This is also called the **dressing**. It helps your incision heal and guards against infection.

Ask your surgeon or surgical team when and how to care for your incision site and change your bandage. Leaving it on too long may delay healing or cause infection.

Contact your surgical team if you develop a fever, or have more drainage than expected from your surgical incision, redness or new swelling at the incision site, or nausea and vomiting that doesn't go away pretty quickly.

Surgical drains

You may have a tube or tubes (called **drains**) coming out of the surgical opening in your skin. Drains pull fluid that collects at the surgery site out of your body. Your surgeon will take them out once they stop draining fluid.

Your surgeon or surgical team will tell you and your caregiver how to care for drains if you have them. You and your caregiver may need to learn how to empty the drain, keep it clean, and change the dressing around the drain.

Eating and drinking

You may not feel like eating or drinking after surgery, but this is an important part of the recovery process. Your surgical team may start you out with ice chips or clear liquids. They will also check to make sure that you are passing urine normally. They may want to measure the amount of urine by having you urinate in a special container.

The digestive tract (stomach and intestines) is one of the last parts of the body to recover from anesthesia and other medicines used during surgery. You'll need to have signs of stomach and bowel activity before you'll be allowed to eat.

Along with checking your surgical wound and other parts of your body, your doctor or nurse will listen for bowel sounds in your belly and will ask if you have passed gas. These are signs that your digestive tract is starting to work normally again. You will probably be on a clear liquid diet until this happens. Once it does, you may get to try solid foods.

Talk with your nurse or registered dietitian about getting the right nutrition. They can also help you return to eating your regular foods.

Physical activity

Being active, such as getting out of bed or even walking soon after surgery, can help prevent problems after surgery. Moving can help speed your recovery by getting your digestive tract working, increasing your circulation, and preventing blood clots and pneumonia.

If you are in the hospital, the nursing staff will help you get out of bed to be sure you are ready to begin physical activity. If you are at home, take your time standing up and moving around. You can increase your activity as you feel up to it.

If you are not able to get out of bed, you might have pneumatic stockings place on your legs. These devices are wrapped around your legs and gently squeeze and release every so often to help your circulation and prevent blood clots.

Your surgical team may also recommend physical therapy and/or occupational therapy to help with your recovery.

Your surgical team will encourage you to do deep breathing exercises. You might be given a device called an incentive spirometer to help you take deep breaths. This helps fully inflate your lungs and reduces the risk of lung infection (pneumonia).

Be sure to let your team know if you're having pain that is affecting your ability to be active so they can give you medicine to control it.

How fast you recover from surgery depends on the kind of surgery you had and your overall health. Be sure to ask your health care team what you might expect right after your surgery.

Going home after surgery

Planning for you to go home or to another setting starts very soon after surgery. Your discharge plan is finalized once you're eating, drinking, and walking. The plan also will depend on other factors, such as the results of your surgery and any tests done afterward.

[Pain control](#)¹ is important as you recover. If you're having pain, be sure to let your health care team know.

Before you are sent home, be sure that you understand these things:

- How to care for your incision (and drains) at home
- What symptoms or side effects you should let your surgeon or surgical team know about right away
- If you have any activity limits (driving, working, lifting, exercising, etc.)
- What you can eat and drink
- What medicines you should take and when you should take them
- Who to call with questions or problems during the day, after hours, or on the weekend.

- Whether you should be doing anything for rehabilitation (physical or occupational therapy)
- When you need to see the doctor again

Everyone recovers differently. Wounds heal at different rates, and some operations are more involved than others.

You may need help at home for a while after surgery. If you need more help than your family members or friends can give you, ask your surgical team if [home care](#)² is an option. They should be able to arrange for a nurse or nurse's aide to visit you at home for a while. This may also depend on your insurance plan.

Adjusting to changes in your body after surgery

Depending on your surgery, you might have changes in how your body looks and works. Some of these changes can be hard to get used to. Feeling upset, sad, or angry about the changes is normal.

It might help you adjust if you have a good idea ahead of time what the likely result of your surgery will be. You will probably still need time to adjust to the changes, even if you knew they were going to happen. Your surgical or cancer care team should be able to help you with those feelings. They won't be surprised if you tell them that you are having a hard time coping with the changes.

Let your surgical or cancer care team know how you are feeling. Be as specific as you need to with your questions, and make sure they give you answers you can understand. It might also help to see a counselor or join a [support group](#)³ for people who have the same kind of cancer or had surgery like yours.

When to call your doctor after cancer surgery

Before you leave the hospital, make sure you know how to contact your surgical team at any time – even after hours, on weekends and holidays.

Some side effects from surgery may come and go quickly, but others might be a sign of a more serious problem. Let your surgical team know right away if you have any of the following symptoms after surgery:

- A fever (ask what temperature you should call for)
- Shaking chills
- Bleeding from your incision or drain site or unexplained bruising and bleeding anywhere else
- Pain or soreness at the surgical site that's getting worse or is not relieved by the pain medicine you were told to take
- Unusual pain anywhere, including in your legs, chest, or belly, or intense headaches
- Nausea and vomiting
- Dizziness
- Shortness of breath or trouble breathing
- Having trouble or pain urinating (peeing); or bloody, bad smelling, or cloudy urine
- Any other symptoms or signs mentioned by your surgical team

Don't hesitate to let your surgical team know about any new problems or concerns you have. It's always best to find out the cause of a problem so it can be treated right away.

Hyperlinks

1. www.cancer.org/cancer/managing-cancer/side-effects/pain.html
2. www.cancer.org/cancer/preparing-for-treatment/home-care.html
3. www.cancer.org/cancer/survivorship/coping/understanding-psychosocial-support-services.html

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Written by

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